

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Go, JavaScript, TypeScript, C++, SQL, Bash

Systems & Performance

Linux profiling, performance optimization, Spark / PySpark, Spark UI, async services, integration testing, pipeline instrumentation / observability

Backend & APIs

Node.js, Express, FastAPI, REST APIs, multi-tenant SaaS, authentication, rate limiting, production debugging

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, git-webhook auto-deploy, Linux configuration

Databases & Caching

PostgreSQL, MySQL, MongoDB, Redis, SQLite

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability and security for **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Monitored production logs and system behavior across feature releases and client deployments, investigated incidents, and shipped fixes for tenant-specific and integration issues; improved operational stability and observability through log monitoring, authentication hardening, and rate limiting.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python

- Cut end-to-end Spark ETL runtime roughly **25%** (17 to 19 min down to 14.3 min) on a **1.4B-row** NYC TLC dataset by replacing shuffle joins with broadcast joins, enabling adaptive query execution and skew-join handling, and tuning shuffle partitions and partitioning.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks; applied column pruning and precomputed fields to cut unnecessary data movement, and documented a repeatable optimization workflow.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x** to **1.74x** via partition pruning, filter pushdown, precomputed derived columns, and adaptive skew-join configuration.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Engineered FastAPI backend and Electron/React frontend to run **fully offline with no external network dependencies**, suited to air-gapped/edge operation; designed async APIs and queues to keep the UI responsive while handling video and audio streams locally on device.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and tool failures; conversation state persists server-side in D1 without an external database.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Hardened a multi-stage agentic web-discovery service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation for review-friendly debuggability.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.